



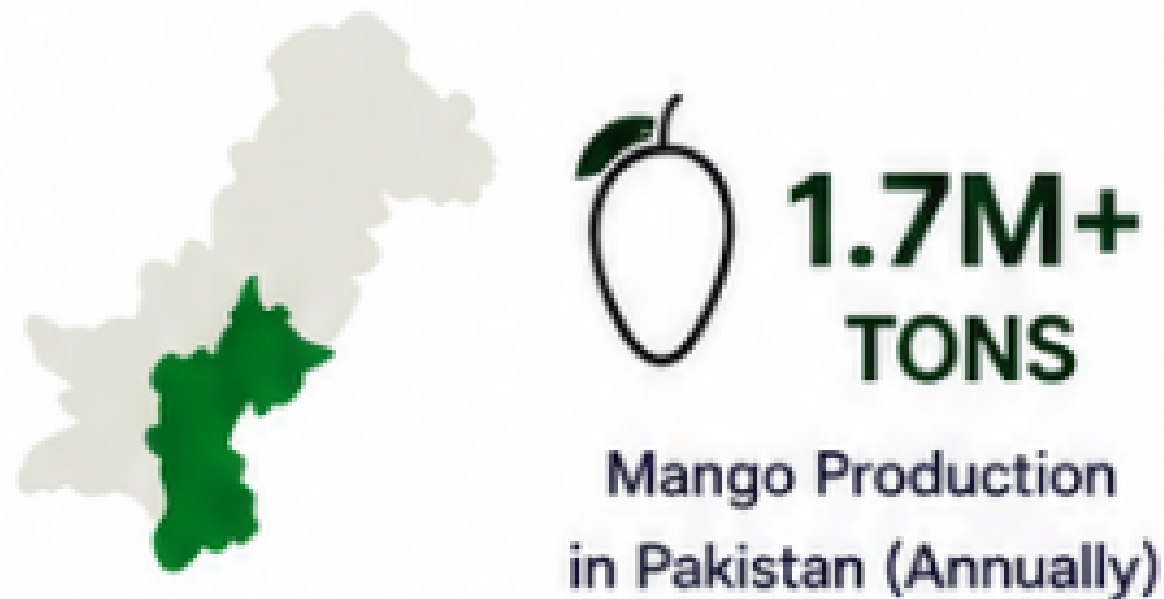
EXPORT EDGE

AI POWERED MANGO SCREENING



1. INTRODUCTION

ExportEdge AI integrates computer vision and AI-based intelligence to improve mango export quality and enable data-driven export decisions.



2. PROBLEM STATEMENT



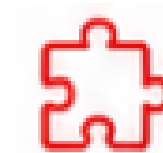
25-40% post-harvest losses due to poor handling



Manual grading causes inconsistent quality classification



Export decisions rely on outdated pricing information



Export decisions rely on outdated pricing information

3. KEY FEATURES



AUTOMATED GRADING



DEFECT DETECTION



FEATURE EXTRACTION

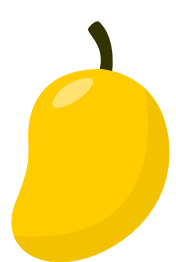


EXPORT INTELLIGENCE

4. OBJECTIVES

- Reduce post-harvest losses by 20% (target)
- Provide real-time export insights
- Improve pricing decisions using AI
- Support ~5,000 exporters and 3,000 facilities

5. WORKFLOW



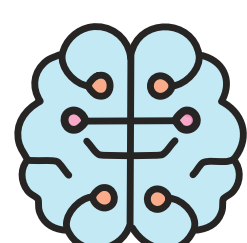
Raw Mango



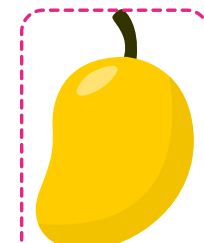
Image Capture



Preprocessing



Classification (Healthy/Diseased)



Segmentation



Feature Extraction



Market Analysis



Report

6. RESULTS

Metric	value
Classification Accuracy (test)	~76%
Segmentation Accuracy (test)	~85%
Inference Time (per image)	~2 sec

Metrics

```
{
  "timestamp": "2026-03-03T14:05:57.342259",
  "image_path": "Mango Variety and Grading Dataset\\Data\\506_833_3",
  "classification": null,
  "defect_analysis": {
    "surface_quality_score": 57.0,
    "color_uniformity_score": 69.67,
    "total_defect_percentage": 1.924,
    "defect_count": 5,
    "dark_spot_count": 2,
    "brown_spot_count": 3,
    "mango_area": 221520.0,
    "total_defect_area": 4262.0,
    "export_grade_impact": "significant",
    "processing_time": 0.2903,
    "defect_regions": [
      {
        "type": "dark_spot",
        "area": 277.0,
        "severity": "moderate",
        "confidence": 0.6893,

```

Extracted Features



Feature Mask

Export Recommendation

Recommended Countries

- United States (US)
- European Union (EU)

Actionable Steps

- ✓ Ensure proper packaging to meet quality standards.
- ✓ Follow labeling and documentation guidelines to avoid any additional processing.

Conditions Before Exporting

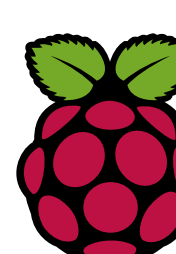
- ✓ No pre-marting conditions are necessary as surface quality is perfect.

Example Report

7. CONCLUSION

ExportEdge transforms traditional mango export workflows into an AI-driven, automated system. It enables accurate quality assessment, reduces post-harvest losses, and helps exporters make data-driven decisions. This solution demonstrates the strong potential of combining computer vision and AI intelligence for real-world agricultural impact.

8. Tools Used



9. SDGs



Advisor: Dr.Shahzad Younis

Co-Advisor: Dr.Momina Moetesum

Team :

M Ammar Bin Akram

M Moosa Sahi

Hannan Yousaf Butt